

Biochar amendment reduced greenhouse gas intensities in the rice-wheat rotation system: six-year field observation and meta-analysis

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ABSTRACT

Biochar amendment to cropland has been recommended as a potential strategy to mitigate climate change. However, estimations of the long-term greenhouse gas (GHG) emissions, namely, the methane (CH₄) and nitrous oxide (N₂O) mitigation potential, from agricultural systems are limited. Here, a six-year field experiment was conducted to simultaneously monitor the CH₄ and N₂O emissions and the yield from a rice-wheat rotation system with nitrogen application (0 and 250 kg ha⁻¹) and biochar incorporation (0, 20 and 40 t ha⁻¹) in southeastern China. The results showed that nitrogen application significantly increased CH₄ and N₂O emissions and the yield compared to the control, while no significant differences were detected among the treatments with no nitrogen. In contrast, biochar amendment resulted in significant decreases in CH₄ and N₂O emissions by 11.2–17.5% and 19.5–26.3%, respectively, with increases in yield by 7.9–9.2%, on average. In addition, considering the GHG emissions, the soil organic carbon sequestration rate and the yield, biochar significantly decreased the net global warming potential and greenhouse gas intensity over the six-year observation period. Our results were further supported by a meta-analysis, which showed that biochar amendment in the field strongly decreased CH₄ and N₂O emissions by 9.3% and 18.7%, respectively, and increased the yield by 9.0% under various complex soil environments. Based on our results, biochar application would increase carbon stock by 10.3% and decrease GHG emissions by 10.4% for Chinese rice and wheat production. Our findings suggest that the application of biochar to cropland is a viable option to combat climate change over the long term by soil carbon sequestration and GHG mitigation.

1. Introduction

Global climate change caused by greenhouse gas (GHG) emissions has attracted unprecedented attention in recent years, and the global warming potential (GWP) of methane (CH₄) and nitrous oxide (N₂O) is 28 and 265 times higher, respectively, than that of carbon dioxide (CO₂) over a hundred year time horizon (IPCC, 2013). Agricultural ecosystems contribute approximately 12% to the total global anthropogenic GHG emissions, as soils account for approximately 37% (mainly as N₂O and CH₄) of agricultural emissions (Paustian et al., 2016; Tubiello et al., 2015). Meeting the growing demand for cropland and grain production will lead to increasing nitrogen fertilizer (N) application and CH₄ and N₂O fluxes from agriculture, effective measures should be taken to abate GHG emissions at high crop productivity (Ma et al., 2013; Xia et al., 2017). The grain growing area accounts for 56.6% of the cropland, and the yield of rice and wheat is more than half

the grain production in China (NBSC, 2018). The rice-wheat rotation system is an important cropping system and is the representative source of GHG emissions in China, which includes the fact that CH₄ fluxes are much larger than the sink contribution in rice cultivation (Jeffery et al., 2016) and that N₂O is mainly generated during wheat seasons while also being produced in rice seasons because of midseason drainage and moist irrigation (Wang et al., 2013). It was estimated that the rice-wheat rotation accounted for 8.2% of the total area for grain production, of which the yield and GHG emissions made up 9.8% and 15.0% from grain fields, respectively (Gao et al., 2018). Therefore, the exploration of strategies to mitigate GHG emissions from rice-wheat rotation systems has important theoretical and realistic significance for both flooded paddy and upland crops.

The addition of biochar to agricultural soil has been recognized for its mitigation potential through long-term carbon (C) sequestration, crop productivity enhancement and GHG emission reductions (Fan

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et al., 2017; Hagemann et al., 2017; Schlesinger et al., 2018). However, previous studies had various conflicting results: Biochar amendment significantly decreased CH_4 release rates by 12% for paddy soils when averaged across all studies (Jeffery et al., 2016; Ji et al., 2018), but some meta-analyses have reported no difference (He et al., 2016) or even an increase (Song et al., 2016) in soil CH_4 emissions induced by biochar. Additionally, the interaction of biochar with the N cycle has long been recognized, meta-analyses of studies have been conducted across a range of cropping systems using various biochar and have shown that biochar application reduced soil N_2O emissions by 30–54% on average (Cayuela et al., 2014; He et al., 2016). Moreover, increased crop productivity responses to biochar amendment also differed in agricultural systems (Biederman and Harpole, 2013; Fan et al., 2017; Li et al., 2017).

The stability of biochar makes it a promising tool to help abate climate change by long-term sequestering of carbon in soils (Bamminger et al., 2018; Wang et al., 2016b). Unfortunately, most previous studies on GHG mitigation strategies only lasted for a short time or were conducted under laboratory conditions (Qin et al., 2016; Verhoeven et al., 2017). Therefore, it is still unclear how biochar influences soil GHG emissions over a long period. For biochar weathering in soil under variable climatic conditions and field management techniques (Verhoeven et al., 2017), the mechanisms and results revealed in short-term studies may not be applicable. Biochar aging in the soil caused changes in the initial properties and effectiveness, for instance, Wang et al. (2016a) found that aged biochar improved soil microbial parameters and contributed to carbon sequestration in soil by negative priming, whereas fresh biochar did not exhibit the same effects. Felber et al. (2014) and Spokas (2013) reported that biochar lost its initial ability to suppress soil N_2O emissions after aging for 1 and 3 years in a field trial, and field-aged biochar stimulated N_2O production from greenhouse vegetable production soils by nitrification and denitrification (Duan et al., 2018). These results suggested that aged biochar may not be suitable for the mitigation of soil N_2O emissions, and studies on the long-term effect of biochar on CH_4 emissions are scarce. Consequently, comprehensive evaluations by long-term field studies are needed to illustrate the persistence of biochar's effect and the mechanisms of GHG emissions under a wide range of agricultural conditions. From the perspective of GHG emissions, the transformation of the radiative forcing of CH_4 and N_2O to the CO_2 equivalent is defined as the GWP (IPCC, 2013). The net exchanges of CO_2 can be measured by soil organic carbon changes (the soil organic carbon sequestration rate, SOCSR), finally, the balance among the net exchange of CO_2 , N_2O and CH_4 constitute the net GWP (NGWP) according to the GWP and the SOCSR (Mosier et al., 2006). In addition, when considering the GHG emissions and grain yield, the NGWP should be expressed comprehensively on a per unit of yield basis (greenhouse gas intensity, GHGI) (Wang et al., 2013).

Here, a six-year field experiment was conducted in the rice-wheat rotation system located in southeastern China to evaluate how yield and GHG (CH_4 and N_2O) emissions were affected by long-term nitrogen application amended with biochar, and then the SOCSR, GWP, NGWP and GHGI were also calculated. Moreover, we presented a comprehensive meta-analysis investigation of the effect of biochar application to soil, with a special focus on field conditions in agricultural systems, on the yield and GHG emissions. The objective of this study was to identify the long-term effect of biochar on the yield and GHG emissions and determine their key controlling factors with respect to flooded rice systems versus upland cropping systems to discuss knowledge gaps for steering future research.

2. Materials and methods

2.1. Experimental site

A six-year field experiment was conducted from June 2012 to May

2018 in Moling town, Nanjing, Jiangsu Province, China (31°52' N, 118°50' E). The field was cultivated under a crop rotation system of rice in summer (June to October) and wheat in winter (November to May). The region is characterized by a typical subtropical climate with an annual average air temperature of 15.7 °C and precipitation of 1050 mm. The soil is classified as Irragic Anthrosols with a silty clay loam texture consisting of 14% clay, 6% sand, and 80% silt, and the main properties of the soil are as follows: bulk density of 1.28 g cm⁻³, pH 5.7, organic C content of 14.6 g kg⁻¹, and total nitrogen content of 1.32 g kg⁻¹ before biochar incorporation.

For the field experiment, biochar was produced from wheat straw at the Sanli New Energy Company in Henan, China, by pyrolysis and thermal decomposition at 500 °C. The biochar had a carbon content of 467 g C kg⁻¹ and a nitrogen content of 5.9 g N kg⁻¹, and the initial values of pH, cation exchange capacity, and ash content were 9.4, 24.1 cmol kg⁻¹ and 20.8%, respectively.

2.2. Design and field management

A total of six treatments with three replicated plots (4 m × 5 m) were established using a randomized block design. Nitrogen fertilizer (urea) was applied at two rates, 0 (N0) and 250 kg N ha⁻¹ crop⁻¹ (N1), for both rice and wheat, and urea was split broadcast at a ratio of 4:3:3 as a basal fertilizer and two topdressings in three fertilizer treatments. Phosphate and potassium fertilizers were applied uniformly as a basal fertilizer to the overall treatments at 60 kg P₂O₅ ha⁻¹ crop⁻¹ and 120 kg K₂O ha⁻¹ crop⁻¹. Biochar was applied at rates of 0 t (B0), 20 t (B1) and 40 t ha⁻¹ (B2) to topsoil plowed at a depth of 20 cm in the field before flooding in June 2012. Only one biochar application was made in this study, as the experiment focused on the long-term effect of biochar. Field management followed local agronomic practices, including cultivation, irrigation, pest and weed control. Table S1 provides the details of the specific schedule of the six rotations from 2012 to 2018.

2.3. Measurements of CH_4 and N_2O fluxes

The CH_4 and N_2O fluxes were measured from June 2012 to May 2018 using a static opaque chamber and gas chromatography over all twelve rice and wheat seasons, following the methods described by Xiong et al. (2015) and Sun et al. (2018a). The top chamber was 0.5 or 1.1 m tall, depending on the plant height, covered a field area of 0.2025 m² (45 cm × 45 cm) and was placed on a fixed PVC base in each plot. To minimize air temperature variations inside the chamber during sampling, the chamber was wrapped with a layer of sponge and aluminum foil. Further details of the chamber configuration were provided in Fig. S1.

For each fluxes measurement, four gas samples were collected from 9:00 to 11:00 am using a 25-mL syringe at 0, 10, 20, and 30 min after the chambers were placed on the fixed base. Over the six-year rice-wheat cycles, fluxes were generally measured once a week, but measurements were conducted once every two days after fertilizer application. The gas samples were analyzed for CH_4 and N_2O concentrations using a gas chromatograph (Agilent 7890A, Shanghai, China) equipped with a hydrogen flame ionization detector (FID) for CH_4 analysis and an electron capture detector (ECD) for N_2O analysis. Argon-methane (5%) and nitrogen gas were used as the carrier gases for ECD and FID, respectively, at a flow rate of 40 mL min⁻¹. The oven and FID were operated at 50 °C and 300 °C, respectively, the temperatures for the column and ECD detector were maintained at 40 °C and 300 °C, respectively. The concentrations of CH_4 and N_2O were quantified by comparing their peak areas with those of standard gases. Four samples of standard gases are regularly adopted at the initial and ending of each measurement.

The fluxes (F) of CH_4 and N_2O were calculated using the following equation:

$$F = \rho \times V/A \times dc/dt \times 273/(273 + T) \times P/1013$$

where F is the fluxes of greenhouse gas ($\text{mg C m}^{-2} \text{h}^{-1}$ for CH_4 ; $\mu\text{g N m}^{-2} \text{h}^{-1}$ for N_2O), ρ is the density of CH_4 (0.5325 g L^{-1}) or N_2O (1.25 g L^{-1}), V is the volume of the static opaque chamber (m^3), A is the cover area of the fixed PVC frame (m^2), dc/dt is calculated using the linear increases in CH_4 or N_2O concentration with time, T is the temperature inside the static opaque chamber ($^{\circ}\text{C}$), P is the air pressure of 1013 (hPa) in the chamber. Sample sets are rejected unless they yield a linear regression value of $R^2 > 0.90$.

The seasonal CH_4 and N_2O emissions were sequentially accumulated from two adjacent measurements.

2.4. Soil measurement and environmental factors

Soil samples were collected from the top layer (0–20 cm) when the field experiment was initiated in June 2012 (only soil organic carbon was measured for every treatment) and the end of the last rice (September 2017) and wheat season (May 2018). Soil samples from each plot were mixed and homogenized, ground to $< 2 \text{ mm}$, and then divided into three aliquots. The first section was air dried to analyze soil organic carbon (SOC), total nitrogen (TN), and pH, the second section was stored at -4°C for analysis of microbial biomass carbon (MBC), and the remaining section was stored at -80°C for molecular analysis. Soil pH was determined in 1:5 (weight:volume) ratios of soil with distilled water using a pH meter (PHS-3C, Shanghai). The SOC was analyzed by wet digestion with $\text{H}_2\text{SO}_4\text{--K}_2\text{Cr}_2\text{O}_7$, the TN was determined using semimicro Kjeldahl digestion using Se, CuSO_4 and K_2SO_4 as catalysts, and MBC was estimated using the chloroform fumigation extraction method.

Soil DNA extraction and quantitative PCR were performed according to Wu et al. (2018), and the abundance of functional genes involved in methanogens and methanotrophs (*mcrA* and *pmoA* genes) was determined in the latest rice season. Ammonia oxidation (AOA and AOB genes), nitrite reductase (*nirK*, *nirS* genes) and N_2O reductase (*nosZ* genes) were simultaneously determined in the last wheat season.

The grain yield was measured following harvest using all the seeds collected at three random positions in each plot using a $1 \text{ m} \times 1 \text{ m}$ frame, and the grain yield of the wheat and rice were determined after the grains were oven dried at approximately 70°C to a constant weight.

Ambient air temperature and precipitation data were provided by the nearby weather station in Nanjing, China.

2.5. Estimation of the SOCSR, GWP, NGWP and GHGI

The soil organic carbon sequestration rate (SOC_{SR}, $\text{t C ha}^{-1} \text{ yr}^{-1}$) was calculated using the following equation (Liu et al., 2015; Xiong et al., 2015):

$$\text{SOC}_{\text{SR}} = (\text{SOC}_t - \text{SOC}_0) \times r \times 20 \times 10^{-1}$$

where SOC_t and SOC_0 are the SOC contents measured at the start and end of the experiment, respectively, and r is the average bulk density (g cm^{-3}). The number 20 represents the thickness of the topsoil.

We used the following equations to calculate the global warming potential (GWP, $\text{t CO}_2\text{-eq ha}^{-1}$), the net global warming potential (NGWP, $\text{t CO}_2\text{-eq ha}^{-1}$) and the greenhouse gas intensity (GHGI, $\text{kg CO}_2\text{-eq kg}^{-1} \text{ grain}$):

$$\text{GWP} = 28 \times \text{CH}_4 (\text{t CH}_4 \text{ ha}^{-1}) + 265 \times \text{N}_2\text{O} (\text{t N}_2\text{O ha}^{-1})$$

$$\text{NGWP} = \text{GWP} - \text{SOC}_{\text{SR}} \times 44/14$$

$$\text{GHGI} = \text{NGWP}/\text{grain yield}$$

where the numbers 28 and 265 represent the IPCC factors without climate-carbon feedback for the conversion of CH_4 and N_2O to CO_2 equivalents, respectively (IPCC, 2013).

The N_2O emissions factor ($\text{N}_2\text{O EF}$, %) was calculated using the

following equation:

$$\text{N}_2\text{O EF} = 100 \times (\text{E}_F - \text{E}_0)/N$$

where E_F (kg N ha^{-1}) is the cumulative N_2O emissions for the N fertilized treatment, E_0 (kg N ha^{-1}) is the cumulative N_2O emissions for the unfertilized treatment (N0), and N is the N fertilizer application rate (kg N ha^{-1}).

2.6. Data processing and statistical analyses

Statistical analyses were performed using JMP 10.0 (SAS Institute Inc., Cary, USA). Two-way factorial analysis of variance (ANOVA) was used to test the effect of nitrogen and biochar amendments on annual CH_4 and N_2O emissions, GWP, SOCSR, NGWP, yield and GHGI. A Tukey's multiple range test was used to determine whether significant differences were found among the treatment means at a significance level of 0.05. The effects of the main driving factors on the yield and GHG emissions were determined using pairwise correlation. Stepwise regressions were used to evaluate the relative importance of soil properties to yield and GHG emissions. Figure preparation was performed by Origin 2017 (OriginLab, USA).

2.7. Meta-analysis

To test the generality of our findings, a comprehensive literature search was conducted of articles that reported CH_4 or N_2O emissions following biochar application in the Web of ScienceTM and China National Knowledge Infrastructure. 'Biochar or charcoal' was used as the keyword, and 'greenhouse gas, methane, nitrous oxide, GHG, CH_4 or N_2O ' was used to refine the search of qualified studies published prior to May 2018. Eventually, data were collected from 60 articles with 209 individual paired measurements by integrating both the control and biochar-amended treatments (Supporting Information 2).

The studies met the following criteria:

- 1 Field studies were continued for at least a full crop season in the field.
- 2 Experiments measured CH_4 or N_2O fluxes and had at least one pair of data (control and treatment).
- 3 Soil and biochar properties, the method of biochar application and experimental durations were clearly described.
- 4 The means, standard deviations/errors, and sample sizes of the variables could be extracted directly from tables, graphs or contexts.

To evaluate the responses of yield and CH_4 and N_2O emissions to biochar application, we calculated the effect sizes based on the natural log-transformed response ratio ($\ln R$) using the mean value of biochar treatment to that in the control:

$$\ln R = \ln (x_t/x_c)$$

where x_t and x_c are the values of the means in the treatment and control groups, respectively. The variance (v) is estimated as:

$$v = \frac{s_t^2}{n_t x_t^2} + \frac{s_c^2}{n_c x_c^2}$$

where n_t and n_c are the sample sizes for the treatment and control groups, s_t and s_c are the standard deviations of the treatment and control groups, respectively.

The categorical random effects model was used to calculate the mean effect size to explain the response of yield and CH_4 and N_2O emissions to soil, biochar properties and other auxiliary variables. The effect sizes and 95% bootstrapped confidence interval (CI) were determined using METAWIN software. When the 95% CI overlapped with zero, biochar application had no significant impact on the variable. The between-group heterogeneity (Q_b) across all data for a given response

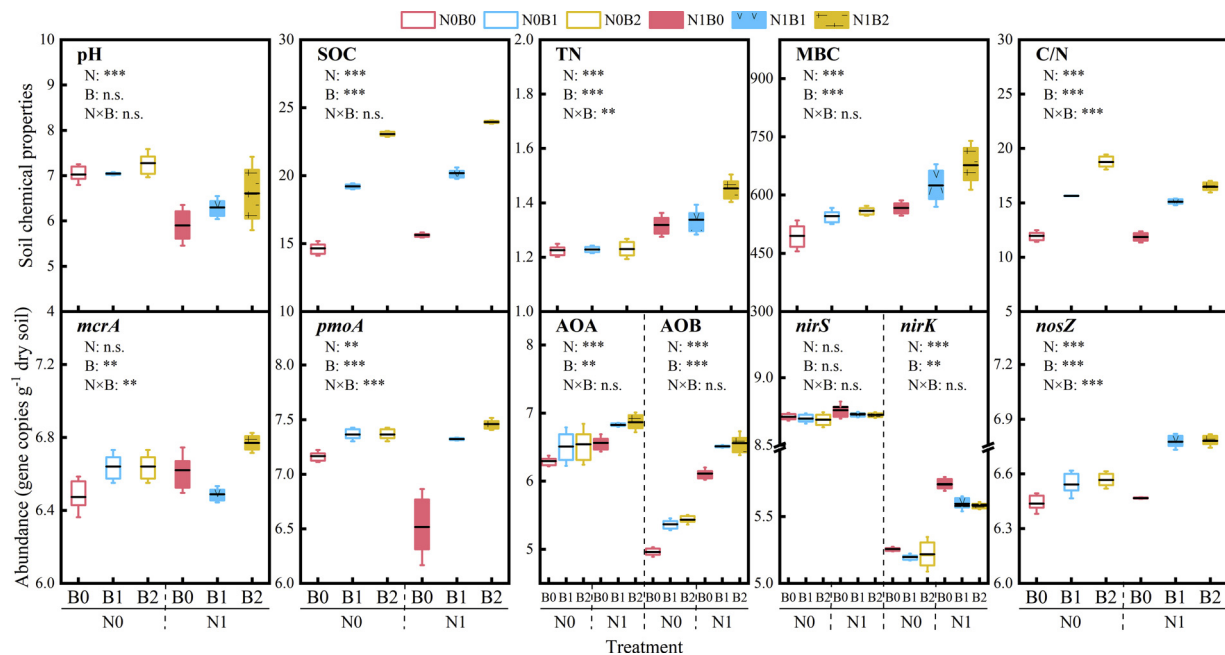


Fig. 1. Variations in soil chemical properties (pH; SOC, g kg⁻¹; TN, g kg⁻¹; MBC, g kg⁻¹; C/N) and the log-10-transformed gene abundances per g dry soil of microbial functional genes (*mcrA*, *pmoA*, *AOA*, *AOB*, *nirS*, *nirK*, *nosZ*) under different treatments over the entire six-year experimental period. The results of two-way analysis of variance on the effect of nitrogen (N), biochar (B), and their interactions (N × B) on soil properties are also shown in the figure. ***Significant at $p < 0.001$; **significant at $p < 0.01$; *significant at $p < 0.05$; n.s. not significant.

variable was calculated to further analyze the biochar effect among different subgrouping categories (Table S4).

3. Results

3.1. Soil chemical properties and abundance of microbial functional genes

The analysis of variance indicated that soil chemical properties strongly depend on nitrogen, biochar incorporation and their interaction (Fig. 1). Compared with the treatments without N addition, N application significantly increased SOC, TN, MBC and C/N at the same biochar amendment rates by 3.8–6.8%, 7.6–18.1%, 14.4–21.0% and 0.8–12.1%, respectively, while N application greatly decreased soil pH by 9.2–15.9%. Biochar and nitrogen amendment significantly increased soil pH, SOC, TN, MBC and C/N by 0.3–11.9%, 29.0–53.1, 0.3–10.2%, 10.2–19.4%, and 27.2–56.9% over those of the treatments without biochar. However, in the treatments without N application, biochar amendment did not affect soil TN, although the soil pH, SOC and C/N increased with the application of biochar. Moreover, increased soil chemical properties in the N1B2 treatment were greater than that in the N1B1 treatment as compared to the treatments with only N addition.

The abundance of microbial functional genes controlling CH₄ and N₂O emissions were shown in Fig. 1. There were no consistent influences in the gene abundances between biochar addition and non-amended treatments when nitrogen was absent. However, application of biochar with nitrogen after six-year significantly increased the *pmoA*, *AOA* and *nosZ* gene abundances by 480.3–698.7%, 150.0–184.0% and 150–180%, respectively, compared to the treatment with nitrogen only, while biochar application significantly decreased *nirK* by 7.5–8.2%. However, no significant difference was found in the *AOA* and *nirS* gene copy numbers among the N1B0, N1B1 and N1B2 treatments.

3.2. CH₄ emissions

All treatments showed similar temporal patterns in the CH₄ fluxes during the six annual rice-wheat rotations (Fig. 2a). The CH₄ emissions increased after transplantation and decreased dramatically at the

midseason drainage, and the CH₄ fluxes increased again to a low emissions peak after reflooding and then decreased gradually to a negligible amount toward harvest. The CH₄ fluxes were significant during the rice seasons but were negligible during the wheat seasons (Table S3; Fig. 2a).

Two-way analysis of variance indicated that annual CH₄ emissions were strongly dependent on the nitrogen, biochar incorporation and their interaction (Table 1). Cumulative CH₄ emissions showed appreciable differences between the treatments with or without N, and biochar amendment independently did not influence the CH₄ emissions in the treatments with no nitrogen. However, biochar application in the rice-wheat rotation system significantly decreased CH₄ emissions over the six-year cycles, averaging 11.2% and 17.5% for N1B1 and N1B2, respectively. In particular, there was no influence on CH₄ emissions when the biochar amendment was plowed into the field at the beginning of the experiment, but over the next five years, the total CH₄ emissions were significantly lower in N1B1 and N1B2 than those in N1B0 (Table S3; Fig. 3a). A multiple stepwise regression analysis showed that CH₄ emissions were controlled by soil pH, followed by MBC, together explaining 81.6% of the variation in CH₄ emissions (Table 2). Moreover, the partial correlation analysis showed that the CH₄ emissions were negatively correlated with the *pmoA* in all treatments (Table S2).

3.3. N₂O emissions

The majority of the N₂O emissions occurred during the wheat growing seasons, peaks were observed following fertilization. The midseason aeration period also stimulated substantial N₂O fluxes in the rice seasons (Table S3; Fig. 2b).

The cumulative N₂O emissions were significantly affected by the application of nitrogen and biochar and their interaction (Table 1). The N₂O fluxes were negligible in the treatments without N, but significant N₂O emissions induced by N application were observed (Table 1, Fig. 2b). A significant reduction in N₂O emissions was found in the biochar treatments in the case of nitrogen fertilization compared with the control. The N₂O fluxes decreased by 19.5% and 26.3% in N1B1 and

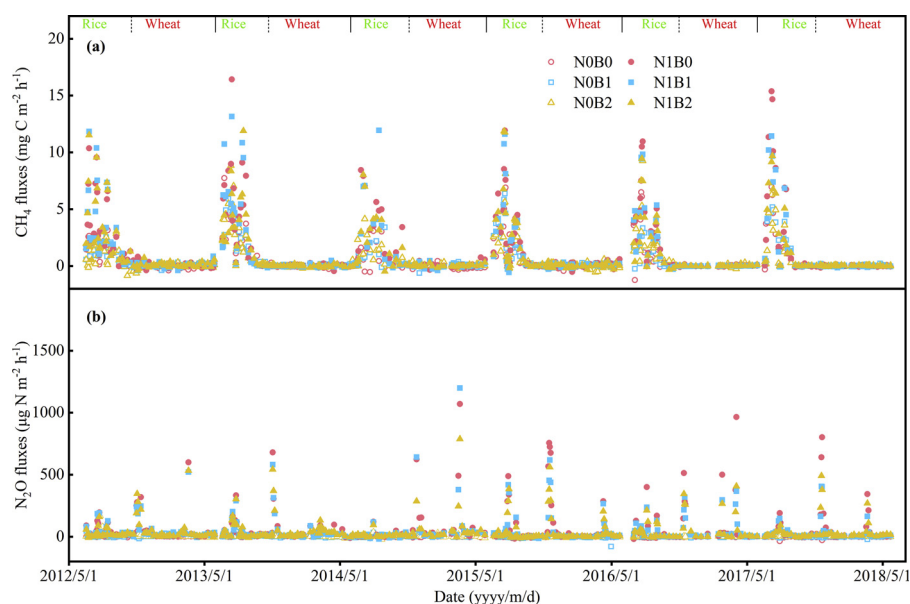


Fig. 2. Fluxes dynamics of CH₄ (a) and N₂O (b) emissions in the rice-wheat rotation system under different treatments during the entire six-year experimental period. Annual seasonal CH₄ and N₂O emissions are shown in Table S3.

N1B2, respectively, with an average of six rice-wheat cultivation periods. In addition, relative decreases in total N₂O emissions by biochar amendment increased with an increasing biochar application rate.

A significant positive linear relationship between the N₂O EF after fertilizer application, and the average temperature and precipitation amounts were found in this study (Fig. S2). Biochar amendment decreased N₂O EF at different temperature and precipitation levels compared to the control. Overall, N₂O emissions were affected by pH and MBC, which explained 88.6% of the variation in N₂O emissions according to the result of multiple stepwise regression analysis (Table 2). A significantly negative relationship was found between N₂O emissions and AOB, AOA and nosZ abundance (Table S2).

3.4. Grain yield

Fig. 3b showed the annual variation in rice and wheat yield over the six-year in this study, showing that the grain yield from rice seasons was greater than that from wheat seasons. N application significantly increased the grain yield on average over the six cycles compared with the treatments without N, and biochar application did not affect the crop yield among the treatments without N. Compared with N1B0, the rice yield was significantly higher by 7.1% and 8.5%, and the wheat yield was higher by 9.2% and 10.4% for N1B1 and N1B2, respectively.

A multiple stepwise regression analysis showed that the grain yield was controlled mainly by soil pH, followed by soil C/N.

3.5. NGWP and GHGI

Over a hundred-year time horizon, the GWP for CH₄ and N₂O was significantly influenced by the nitrogen and biochar application (Table 1), which decreased markedly by 12.1% and 18.4% for N1B1 and N1B2, respectively, relative to N1B0 on average, and high levels of biochar application showed a greater reduction in the GWP, while no significant differences among N0B0, N0B1 and N0B2 were observed. Moreover, CH₄ emissions accounted for more than 80% of the total GWP average over the entire period (Table S3; Fig. 3a). The SOCSR ranged from 0.03 t C yr⁻¹ for N0B2 to 0.46 t C yr⁻¹ for N1B2 (Table 1), which tended to increase in N1B1 and N1B2 compared with N1B0 but was not significant in the treatments with no nitrogen.

The NGWP and GHGI strongly depended on the nitrogen application and biochar incorporation (Table 1). No significant difference in the NGWP was observed among the N0B0, N0B1 and N0B2 treatments, whereas the average NGWP for six years was 29.4% and 50.5% lower for the N1B1 and N1B2 treatments, respectively, relative to the N1B0 treatment when nitrogen was amended. Similarly, no significant difference in the annual GHGI was found for treatment without

Table 1

Mean seasonal CH₄ and N₂O emissions, GWP, SOCSR, NGWP, grain yield and GHGI under different treatments over a six-year experimental period.

Treatment	CH ₄ kg C ha ⁻¹	N ₂ O kg N ha ⁻¹	GWP t CO ₂ -eq ha ⁻¹	SOCSR t C ha ⁻¹	NGWP t CO ₂ -eq ha ⁻¹	Grain yield t ha ⁻¹	GHGI kg CO ₂ -eq kg ⁻¹ grain
N0B0	38.69 ± 2.65c	0.18 ± 0.04c	1.52 ± 0.08c	0.06 ± 0.1b	1.31 ± 0.34c	5.89 ± 0.14c	0.22 ± 0.06b
N0B1	39.87 ± 0.81c	0.23 ± 0.04c	1.57 ± 0.05c	0.07 ± 0.08b	1.33 ± 0.33c	5.76 ± 0.12c	0.25 ± 0.07b
N0B2	37.71 ± 5.23c	0.20 ± 0.04c	1.46 ± 0.29c	0.03 ± 0.07b	1.37 ± 0.48c	5.77 ± 0.28c	0.25 ± 0.07b
N1B0	83.26 ± 2.52a	3.53 ± 0.24a	4.56 ± 0.17a	0.12 ± 0.06b	4.12 ± 0.07a	10.94 ± 0.35b	0.39 ± 0.02b
N1B1	73.94 ± 0.92b	2.84 ± 0.07b	4.01 ± 0.06b	0.30 ± 0.14ab	2.91 ± 0.56b	11.80 ± 0.49ab	0.25 ± 0.04ab
N1B2	68.73 ± 3.54b	2.60 ± 0.21b	3.72 ± 0.16b	0.46 ± 0.16a	2.04 ± 0.63bc	11.95 ± 0.57a	0.17 ± 0.05a
ANOVA results							
N	***	***	***	***	***	***	n.s.
B	***	***	***	*	***	n.s.	**
N × B	**	***	***	**	***	*	***

N0: without N fertilization, N1: 250 kg N ha⁻¹ urea amendment; B0: without biochar amendment, B1: 20 t ha⁻¹ biochar amendment, B2: 40 t ha⁻¹ biochar amendment. Means ± SD with different letters in the same column indicate significant differences according to Tukey's multiple range test ($P < 0.05$) among all treatments. ***Significant at $P < 0.001$; **significant at $P < 0.01$; *significant at $P < 0.05$; n.s. not significant.

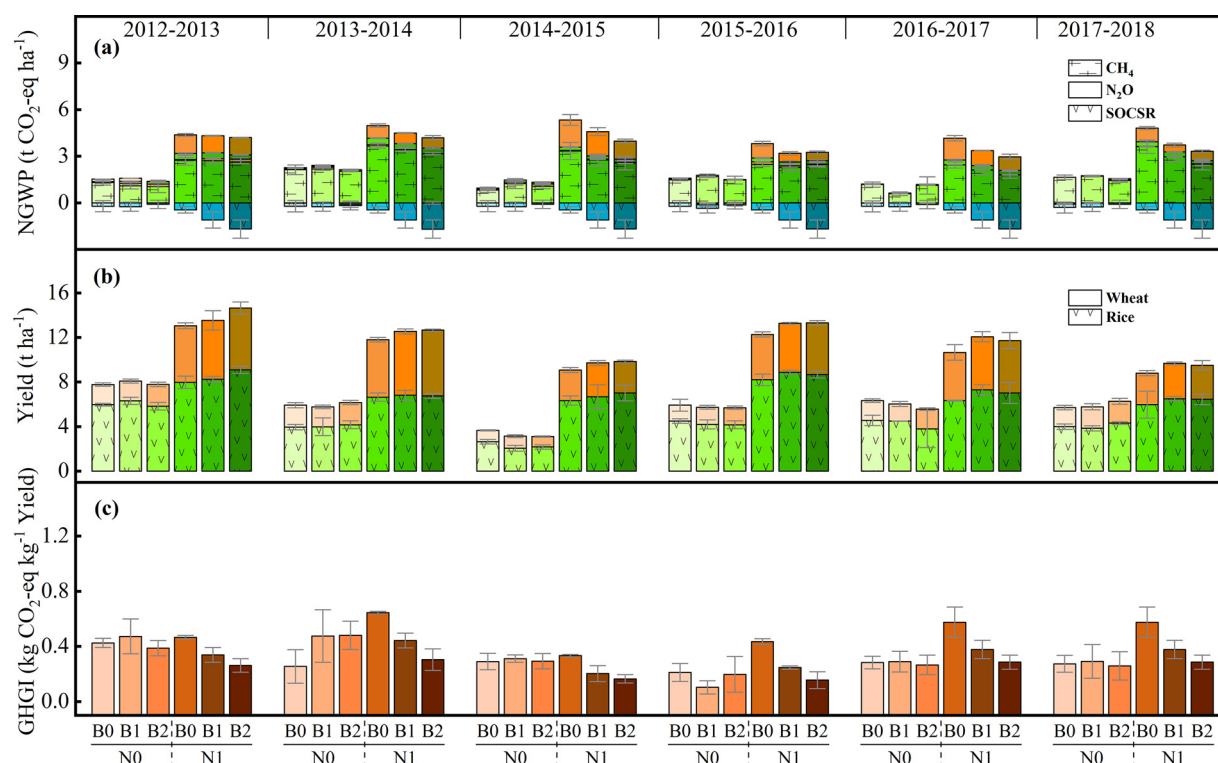


Fig. 3. Net global warming potential (a), yield (b), greenhouse gas intensity (c) in the rice-wheat rotation system under different treatments during the entire experimental period. The bars indicate the standard error of the mean for the three replicates of each treatment.

Table 2

Stepwise regression models between yield, CH₄, N₂O emissions and soil properties.

Dependent variables	Regression equation	p	R ²
Yield	Yield = 0.812-0.633pH + 0.523C/N	< 0.001	0.654
CH ₄ emissions	CH ₄ = -0.729pH + 0.371MBC	< 0.001	0.816
N ₂ O emissions	N ₂ O = -0.56pH + 0.369MBC	< 0.001	0.886

pH: soil pH; C/N: soil C/N ratio; MBC: soil microbial biomass carbon.

nitrogen, although the annual GHGI significantly decreased by 35.9% and 56.4% for N1B1 and N1B2, respectively, among the six periods of rice-wheat cultivation.

3.6. Results of the meta-analysis

On average, the yield and CH₄ and N₂O emissions in the control treatment were 7.7 t ha⁻¹, 174.6 kg C ha⁻¹ (only for paddy), and 2.7 kg N ha⁻¹ among the meta-analyses, respectively (Supporting Information 2). Biochar application significantly increased the yield by 9.0%, while it decreased the soil CH₄ (only for paddy) and N₂O fluxes by 9.3% and 18.7%, respectively (Fig. 4). The soil and biochar characteristics and the experimental conditions were significantly correlated with the response ratios of the yield and soil GHG fluxes across all the studies (Table S4, Fig. S3, 4, 5). The addition of biochar with different properties generally improved the crop yield in various soils, and biochar with a lower C/N was more effective at promoting productivity in clay soil with a higher N application. For CH₄ fluxes, greater negative effects occurred in lower pH and C/N soils when biochar with a higher C/N was amended. Soil texture, soil organic C and soil C/N were shown to be key factors influencing N₂O emissions, while biochar with a higher pyrolysis temperature exerted a greater mitigation of N₂O emissions in higher C/N soil with a higher N application (Fig. S6).

Moreover, the long-term effects of biochar amendment on the yield and GHG fluxes were confirmed by our meta-analysis (Fig. 4). The yield

enhancement and reduction in N₂O emissions in biochar-amended soil persisted over the studied years after a single amendment in field experiments. Biochar application did not change soil CH₄ fluxes in field studies for a shorter period (< 0.5 year) but significantly decreased soil CH₄ fluxes by 18.0-50.1% in experiments sustained over several years.

4. Discussion

4.1. Effects of biochar amendment on grain yield

The grain yield was strongly affected by N application in the field trial (Table 1; Fig. 2b). N fertilization significantly increased the total yield compared to the treatments without N, which was in agreement with previous results suggesting that nitrogen was the most decisive factor for increasing agricultural productivity (Biederman and Harpole, 2013; Li et al., 2017; Sun et al., 2018b). Increases in the soil SOC and TN contents induced by N application may explain the higher crop yield during the experimental period (Fig. 1). Our data also showed that biochar and nitrogen addition could enhance the crop yield (Fig. 2b), which could be explained by improved soil biochemical properties and the stimulation of microelements by biochar incorporation. According to the results of multiple stepwise regression analysis, the grain yield was controlled by the soil pH and soil C/N (Table 2). Although N application resulted in a lower soil pH than that of the corresponding treatment without N, with an increase in soil pH by biochar under N application, the available mineral nutrient supply could be enhanced for plant growth (Biederman and Harpole, 2013; Zhao et al., 2015). Biochar amendment significantly increased the SOC by 26.9–45.8%, as supporting by Wang et al. (2016b), who reported that the remaining 97% of biochar contributed directly to long-term C sequestration in soil. The higher SOCSR in biochar treatment with nitrogen amendment (Table 1) suggested that biochar facilitated interaction with soil minerals and increased the soil C/N. Thus, applying biochar to soil has benefits for enhancing food production over the long term. As further elucidated by the meta-analysis, the amendment of biochar with high

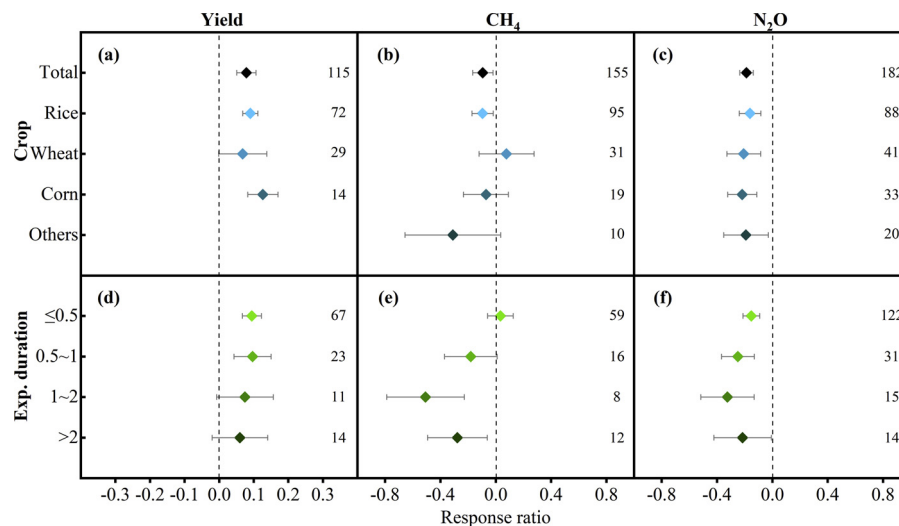


Fig. 4. The effect of biochar application on yield (a), CH₄ (b), and N₂O (c) emissions differed with crop type and experimental duration. Dots with error bars denote the overall mean response ratio and the 95% CIs.

pH and C/N increased crop productivity by 9.0% on average for rice and upland crops (Fig. 4a).

Generally, no significant differences in responses were observed in experiments sustained over several years and in those conducted for a shorter period according to field and meta-analysis (Fig. 3b; Fig. 4d), the increased crop yield response to biochar amendment could be persistent in the long term. We observed that biochar significantly increased soil pH, TN, TC, MBC and C/N after six-year of amendment (Table 1), which was consistent for improving soil quality and increasing the bioavailability of nutrients for crop growth.

4.2. Effects of biochar amendment on CH₄ emissions

In this six-year field study, we found that annual CH₄ emissions were strongly dependent on nitrogen incorporation (Table 1). The production of CH₄ is primarily determined by substrate availability, and N fertilizer stimulated CH₄ emissions by enhancing the C substrates for methanogenic archaea from enhanced rice biomass (Ma et al., 2018; Sun et al., 2018a,b). Nevertheless, the addition of biochar to flooded paddies increased the substrate availability through providing a higher biomass and a labile C pool of biochar (Wang et al., 2012). Therefore, previous studies showed that biochar has no significant effect or that it increased (Wang et al., 2012; Zhang et al., 2010) CH₄ emissions from paddies, concluding that biochar amendment is not suitable for the mitigation of CH₄ emissions. There were no significant differences between biochar and control treatments when nitrogen was amended considering only the first rice growing season, however, biochar significantly reduced CH₄ emissions during the following five rice seasons (Table S3; Fig. 2a). Although the fast mineralization of the labile biochar compounds and the biochar-induced priming effects on soil organic matter could exist in the short term (Maestrini et al., 2015; Wang et al., 2016b), biochar and nitrogen amendments contributed directly to long-term C sequestration in soil and markedly inhibited the substrate availability for CH₄ production, which was demonstrated by the increased SOC and SOCSR in rice paddy soils (Table 1; Fig. 2a). We believe that biochar amendment has advantages of carbon sequestration and decreasing substrate availability for CH₄ emissions.

The CH₄ fluxes were a net consequence of production, oxidation and transport from the soil (Cai et al., 2007), and CH₄ emissions were also determined by the balance between the activities of methanogens and methanotrophs (Feng et al., 2012; Singh et al., 2017). Methanotrophs were regarded as a natural approach to address the greenhouse effect, as they consumed up to 40 Tg CH₄ year⁻¹ and sequestered more than

50% of the methane produced in soils (IPCC, 2013). The increased methane oxidation rate plays a pivotal environmental role in mitigating climate change. Our results showed that lower CH₄ emissions occurred in N1B1 and N1B2 than in N1B0, which was mainly driven by the higher pmoA in the biochar amendment (Fig. 1). In line with Wu et al. (2018), who found a higher pmoA/mcrA ratio after three years of biochar amendment compared to the control throughout different rice developmental stages, a potential explanation was that the methanotrophic communities were more sensitive than methanogens to rising soil pH, which was supported by the positive correlation between pH and the abundance of pmoA (Table S2). Similarly, increased CH₄ oxidation activity by biochar was revealed in previous studies (Feng et al., 2012; Qin et al., 2016). Methane oxidation in soil was also shown to be controlled by oxygen and nutrient supply (Jiang et al., 2017; Singh and Seneviratne, 2017). The dissolved oxygen concentration in paddy soil was the key factor in promoting CH₄ oxidation (Ma et al., 2018), and higher yield and biomass of the biochar treatment facilitated O₂ transport into the rhizosphere and stimulated CH₄ oxidation. Xu et al. (2016) also found that biochar reduced the soil profile CH₄ concentrations in the primary CH₄ production sites during the rice seasons at the same location. Simultaneously, nitrogen acts as a regulatory factor of methane oxidation in agricultural soils. Hu et al. (2015) suggested that nitrate is probably a better N source for methanotrophs in rice field soil and that soil with biochar amendment along with a high activity of AOB and AOA (Fig. 1) could generate more nitrate to serve as a substrate for CH₄ oxidation in the rhizosphere. All results suggested that an increase in methanotrophs by biochar amendment could persist in our paddy field after six-year of amendment.

Although CH₄ fluxes showed great interannual variations along with weather pattern and field management throughout the experimental period, by studying six consecutive years of CH₄ emissions, we revealed a significant reduction in CH₄ emissions by 11.2% for N1B1 and 17.5% for N1B2 (Table 1). The generality of our findings was further confirmed by our meta-analysis for paddy fields, in which biochar application significantly decreased soil CH₄ emissions by 9.3% (Fig. 4b). Similar to our field studies, biochar application did not change soil CH₄ fluxes for a shorter period (< 0.5 year) in the meta-analysis, but greater negative responses were observed in experiments sustained over several years (Fig. 4e). The uncertainty of shorter studies prompted us to improve understanding on the long-term effect of biochar under field conditions. These data suggested that the effect of biochar on CH₄ emissions may be generalized under a wide range of field conditions. Our study confirmed that incorporating biochar has a long-term effect

on CH₄ mitigation.

4.3. Effects of biochar amendment on N₂O emissions

Overall, substantial N₂O emissions only occurred for the N treatment following N fertilization in the wheat season (mainly) or drainage practices in the rice season (slightly) throughout the six-year experimental period (Fig. 2b). N application was the most decisive factor for N₂O emissions due to negligible N₂O fluxes in the treatments without N (Table 1). Because N application rates and other management practices were the same for every treatment with N, the annual variations in N₂O emissions might be explained by the environmental conditions, including temperature and rainfall. The N₂O EF values are important for evaluating the loss of input N fertilizer as N₂O gas (Ma et al., 2018). We observed a significant positive relationship between the N₂O EF and temperature or precipitation for every N application in the six wheat growing seasons (Fig. S2), and biochar treatment was lower than N treatment alone under complex environmental conditions. The results were also supported by our meta-analysis (Fig. 4c) and by Verhoeven et al. (2017), who reported that biochar addition to soil decreased N₂O emissions by 18.7% (a total of 182 observations from 54 studies) and 12.4% (a total of 122 observations from 40 studies) under field conditions. The response ratio of soil N₂O emissions to biochar was significantly negatively correlated with the nitrogen application rate (Fig. S6), suggesting that the mitigation potential of biochar amendment was preferable in high N application agricultural ecosystems.

Our results showed that biochar and nitrogen amendment significantly increased the abundance of AOA, AOB and nosZ, whereas it decreased the abundance of nirS and nirK relative to the treatment with the only N amendment (Fig. 1). Therefore, the primary reason for N₂O mitigation appeared to be the significant changes in the activity of the nitrifying and denitrifying microorganisms. The increases in the nosZ abundance suggested that biochar could decrease soil N₂O emissions by enhancing its further reduction to N₂, which is in agreement with many previous results (Cayuela et al., 2013; Duan et al., 2018). The reduction in nitrite reductase nirS and nirK genes in the biochar treatment also indicated the inhibitory effect of biochar on denitrification, and biochar amendment could decrease N₂O emissions from denitrification. Biochar may influence different N₂O-producing processes by altering soil aeration, N availability, soil pH or sorption of labile C, leading to shifts in the microbial community (Clough et al., 2013). We found that the pH was a key factor in influencing emissions of N₂O (Table 2), and the liming effect of biochar is able to facilitate the transfer of electrons to denitrifying microorganisms in soil under optimum denitrification conditions (Cayuela et al., 2013). Based on its porous structure and large surface area, enhanced soil aeration by biochar could inhibit N₂O emissions from denitrification pathway (Fan et al., 2017). Simultaneously, biochar amendment significantly decreased the diffusive fluxes of N₂O in the rice-wheat annual rotation (Zhou et al., 2016). Therefore, the underlying mechanism of how biochar affects those processes needs to be illustrated in future research. The increased abundances of AOA and AOB suggested that biochar accelerated soil N₂O produced by the elevated nitrification pathway (Duan et al., 2018), and the enhancement of nitrification also indicated the greater transformation rate of ammonium to nitrate. Therefore, we speculated that biochar acceleration of soil N₂O emissions via the nitrification process only occurred in the early stage after fertilization. Then, the N₂O emissions were limited due to the higher soil nitrate concentrations and lower denitrification potential in the biochar plot in the subsequent crop-growing stage, the higher N₂O emissions only occurred under higher ammonium accumulation in the treatment without biochar. Therefore, increased nitrification did not promote N₂O emissions in our field experiment, and biochar decreased soil N₂O emissions by promoting the transformation of ammonium to nitrate and inhibiting denitrification. Moreover, Dong et al. (2017) found that N1B1 and N1B2 increased the nitrogen use efficiency by 30.1% and 35.9%, respectively, and the relative increases

in nitrogen use efficiency and yield induced by biochar amendments were the potential factors to decrease the N loss by N₂O emissions.

Although the majority of N₂O emissions occurred during the wheat-growing season in the rice-wheat rotation system, the rice season contributed an average of 25.0% of annual N₂O emissions from the N-fertilized treatments (Table S3). Therefore, our long-term field measurements highlight that N₂O emissions cannot be ignored from the subtropical rice paddy with mid-drainage. In contrast to the wheat season and meta-analysis, biochar has no significant effects in rice seasons across the six-year of data (Table S3). The peak N₂O emissions occurred in mid-drainage under alternate wetting-drying, and high moisture was associated with enhanced nitrification and coupled nitrification-denitrification (Verhoeven et al., 2018). For one week surrounding the onset of alternate wetting-drying, biochar treatment more likely enhanced nitrification in the upper soil layers and the related N₂O emissions in the short term. Fortunately, the N₂O emissions from rice seasons were negligible because they contributed only 4.6 % to the total GWP on average (Fig. 3a).

Here, we show that biochar amendment produced long-term effects on N₂O emissions through a field study for six years and a meta-analysis, although only 15.9% of all observations reported data from multiple years in the meta-analysis (Supporting Information 2). To our knowledge, few studies have assessed the long-term N₂O suppression potential by investigating the impact of biochar from long-term field experiments, including decreased (Felber et al., 2014), lost (Duan et al., 2018; Spokas, 2013) or unchanged (Hagemann et al., 2016) initial ability of field-aged biochar to mitigate N₂O from microcosms and field studies. However, Hagemann et al. (2016a) obtained aged biochar by storing it in black plastic bags for three years, the characteristics of naturally weathered biochar might differ from those of particles derived from the field due to the complex soil environment. Spokas (2013) and Duan et al. (2018) separated aged biochar from the soil through sieving, and they only focused on the properties of aged biochar while ignoring the biochar-rich soil. The amendment of soil with biochar as a promising solution to regulate the soil N cycle and reduce N fluxes not only depended on biochar characteristics but also was related to the alteration of soil physicochemical properties and soil microbial communities in the long term. In the current study, biochar amendment six-year ago still obviously increased AOB and nosZ abundances, indicating the long-term potential effect on soil nitrogen transformation. It is thus important to characterize the longevity of the impact of biochar on N₂O production for different biochars and soils in field studies, thereby optimizing biochar-soil combinations for reducing soil N₂O emissions.

4.4. Effects of biochar amendment on the NGWP and GHGI

Here, we showed that the GWP of CH₄ and N₂O was decreased markedly with biochar treatment when nitrogen was amended. Considering that the GWP is dominated by CH₄ emissions in the rice-wheat rotation system (Table S3; Fig. 3a), efforts to reduce the GWP in the rice-wheat rotation system should focus on CH₄ emissions in the rice growing season. Comprehensively considering the emissions of CH₄ and N₂O and the SOCSR in the rice-wheat cropping system, we found that biochar and nitrogen application significantly decreased the NGWP due to decreased CH₄ and N₂O emissions and increased the SOCSR in the biochar treatment (Fig. 3a). In contrast, the grain yield for biochar treatments was significantly greater than that for the control. The NGWP should also be evaluated as a function of crop yield (GHGI) to achieve the dual goal of mitigating CH₄ and N₂O emissions while guaranteeing crop yields. Consistent with the NGWP, biochar application significantly decreased the seasonal and annual GHGI compared with the no biochar treatment when N was applied (Fig. 3c). Therefore, biochar played a specific role in mitigating climate change over the long term.

Overall, the per unit emission values were averaged to be 174.6 kg C ha⁻¹ for CH₄ and 2.7 kg N ha⁻¹ for N₂O (Supporting Information 2),

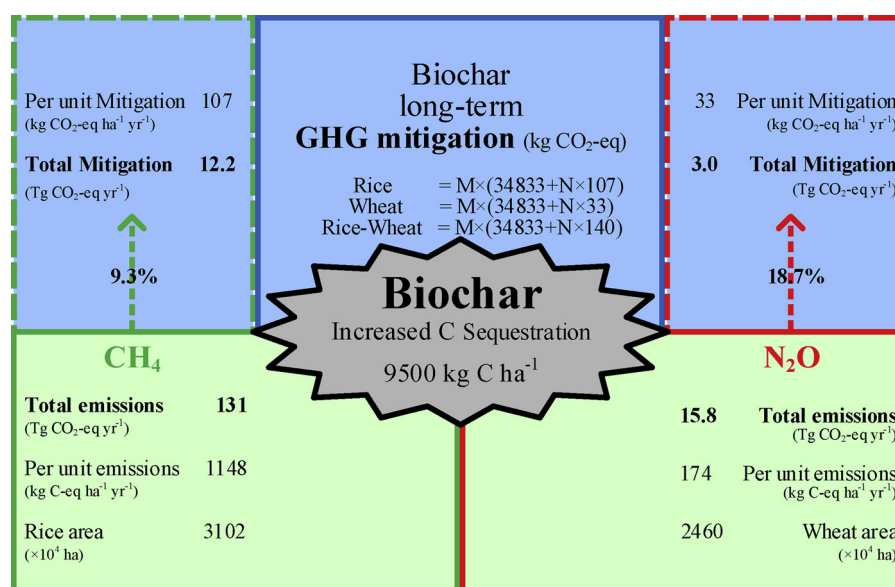


Fig. 5. Overview of soil carbon sequestration and greenhouse gas mitigation by biochar amendment in rice and wheat fields in China. The details of the calculation are shown in Table S5.

which is in good agreement with the values of 183 kg C ha⁻¹ and 2.9 kg N ha⁻¹ estimated by Gao et al. (2018) in China. The CH₄ and N₂O emissions in the Chinese paddy and wheat field were 131 and 15.8 Tg CO₂-eq every year, and 15.2 Tg CO₂ (10.4% of total emissions) could potentially be mitigated by biochar applied to the field according to our estimate (Table S5; Fig. 5). In addition, Tang et al. (2018) estimated that the C pool of cropland for China was 92.1 t ha⁻¹, and biochar application would increase carbon stock by 10.3% immediately based on our results.

Our long-term field investigation (six-year) showed that the application of biochar to rice-wheat rotation systems significantly reduced GHG emissions and increased the crop yield, resulting in a decreased NGWP and GHGI compared with traditional cultivation practices. This result was primarily attributed to long-term comprehensive changes in soil properties following biochar addition, including increased soil pH and C/N, which ameliorated the soil environment to stimulate CH₄ oxidation and N cycling processes. In addition, CH₄ emissions were the dominant contributor to the GWP from the rice-wheat rotation system throughout the experimental period, and efforts to reduce the GWP should focus on CH₄ in paddy fields. The meta-analysis aimed at field studies further suggested that biochar application appeared to be a good strategy to mitigate global warming and increase the yield in fertilized regions over a long period on a global scale. As a result, it could be concluded that the application of biochar to cropland is a viable option to abate climate change in the long term by soil carbon sequestration and GHG mitigation. Therefore, to elucidate the sustainable effect of biochar on soil GHG fluxes, field experiments performed on historically charcoal-rich soils should be broadened to a wider range of environmental and management factors.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.agrformet.2019.107625>.

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